ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) | 979-8-3503-4485-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICASSP48485.2024.10446966

AN EXPLICIT MULTI-MODAL FUSION METHOD FOR SIGN LANGUAGE TRANSLATION

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ABSTRACT

Sign Language Translation (SLT) aims to convert sign language videos into corresponding spoken text sequences. However, the inherent modality gap between sign language video and text hinders the development of SLT. Motivated by the linguistic consistency between gloss[1] and text, we propose EMF-SLT, an **E**xplicit **M**ulti-modal **F**usion method for **S**ign **L**anguage **T**ranslation to mitigate the modality gap with the help of gloss. Specifically, EMF-SLT first leverages a vector quantizer and a fusion module to align and fuse sign language and gloss features, respectively, resulting in more informative multi-modal features for the decoder. Then, a multi-task mutual learning framework is introduced to regularize the output predictions from different modalities, which ensures the consistency of outputs across modalities and encourages different modalities to learn from each other. Experiments on two SLT benchmarks and further analyses show that our method achieves significant improvements over the baselines and effectively alleviates the modality gap.

***Index Terms*—** Sign language translation, low-resource, multi-modal fusion

1. INTRODUCTION

Sign Language Translation (SLT) is a sophisticated cross-modal task that converts sign language videos into spoken text, making it easier for hearing-impaired people to communicate with non-sign language speakers [1, 2]. The traditional method decomposes the SLT into a cascaded system [1, 3], consisting of two modules: Continuous Sign Language Recognition (CSLR) and Gloss to Text Translation (G2T), which first obtains the predicted gloss tokens and then goes on to do the translation. However, this method greatly suffers from error propagation and high latency. To break this bottleneck, recent work has focused on a unified end-to-end paradigm [4, 5, 6, 7, 8, 9, 10, 11], aka Sign2Text (S2T), which uses a single model to directly learn the mapping relationship between the sign language input and the target text, and its translation quality generally outperforms the traditional cascade system.

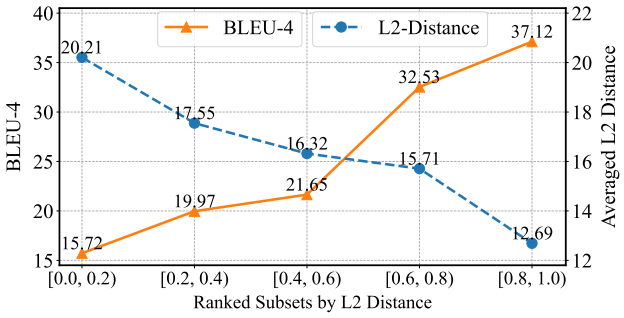

Fig. 1. The relationship between the modality gap and the translation quality. The PHOENIX14T dev set is divided into 5 equal-sized subsets based on the L2 distance of the sentence-level representations between sign language and the text generated by recognized baseline Joint-SLRT [4].

The primary challenge in training an effective end-to-end SLT model stems from the inherent modality gap between sign and text [4, 5, 8]. As shown in Fig 1, a significant improvement in translation quality can be observed as the L2 distance decreases, which means that the modality gap has a significant effect on the translation quality. Existing methods typically introduce the gloss as an intermediate supervised signal, attempting to implicitly model different modalities through a multi-task manner, achieving impressive improvements in SLT task. However, they often ignore the linguistic consistency between gloss and text due to the obvious overlap of tokens [12], especially in Chinese sign language, resulting in a failure to bridge the modality gap effectively.

Based on these considerations and inspired by recent studies on cross-modal task [13, 14, 15, 16], we propose an explicit multi-modal fusion method for SLT (EMF-SLT), to reduce the modality gap between sign and text. Concretely, EMF-SLT first employs a vector quantizer to align gloss and sign representations into a shared discrete space. Due to the shared vocabulary between gloss and text in our method, this alignment serves as a crucial bridge to mitigate the modality gap between sign and text. A fusion module

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[1]Gloss is a verbatim transcription of sign language.

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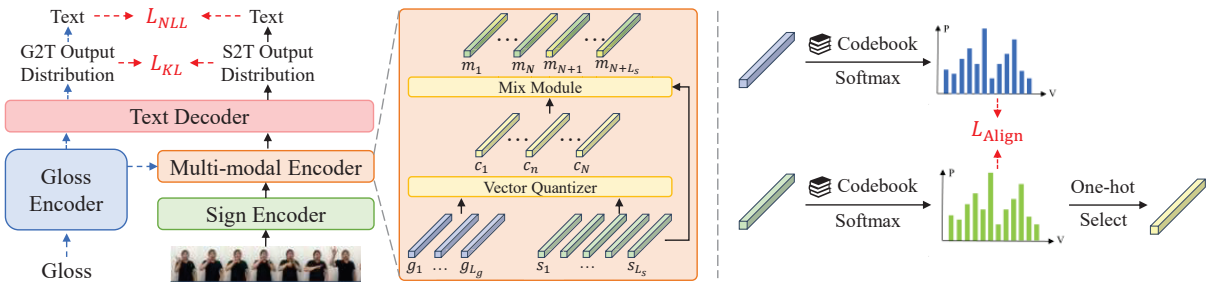

(a) Model architecture

(b) Calculation process of the vector quantizer

Fig. 2. An overview of the proposed EMF-SLT model.

is then introduced to fuse the sign representation and gloss-like representation (the output of vector quantizer). This facilitates multi-modal semantic interaction during training and inference, even though gloss is unavailable for inference in practice. Furthermore, EMF-SLT concurrently accepts gloss and multi-modal representations as input to the text decoder and regularizes their predictions through a multi-task mutual learning framework. This framework ensures the consistency of outputs across modalities and encourages different modalities to learn from each other, thus alleviating the modality gap. Finally, experimental results show that EMF-SLT achieves significant performance improvements over the baselines on two SLT benchmarks.

2. APPROACH

As depicted in Fig 2(a), EMF-SLT includes four sub-modules: a sign encoder, a gloss encoder, a multi-modal encoder, and a text decoder. We utilize the SMKD model [17] pre-trained on the CSLR task as the sign encoder and freeze it during SLT training. The gloss encoder and text decoder are composed of three Transformer [18] encoder and decoder layers, respectively. The multi-modal encoder introduces a vector quantizer and a fusion module to align and fuse the sign and gloss features. Its elaborate description is provided below.

2.1. Multi-modal Encoder

The multi-modal encoder is composed of two sub-modules: vector quantizer and fusion module as illustrated in Fig 2(a).

Vector Quantizer. To effectively fuse the features across modalities, we introduce a vector quantizer to align sign and gloss features in a discrete space. However, it is challenging to perform cross-modal alignment due to the inconsistent lengths between sign and gloss features. Inspired by the Q-former [16], given the sign features $\mathbf{S} = (s_1, \ldots, s_{L_s})$ and gloss features $\mathbf{G} = (g_1, \ldots, g_{L_g})$ respectively output by sign and gloss encoders, we introduce a learnable query $\mathbf{Q} \in \mathbb{R}^{N \times d}$ and utilize a cross-attention operation to transform these two features into consistent-length features, $\hat{\mathbf{S}} = (\hat{s}_1, \ldots, \hat{s}_N)$ and $\hat{\mathbf{G}} = (\hat{g}_1, \ldots, \hat{g}_N)$.

We then discretize these two features into a shared codebook which is essentially a vocabulary consisting of V virtual tokens. In order to align the representations of different modalities in discrete space, we design an alignment loss for vector quantizer to select the same tokens codebook entries for the sign features $\hat{\mathbf{S}}$ and the corresponding gloss features $\hat{\mathbf{G}}$. To be specific, we consider the distribution of the gloss feature across the codebook as the target to supervise the vector quantizer to produce the same distribution for the sign feature, as shown in Fig 2(b). The alignment loss is defined as:

$$\mathcal{L}_{\text{Align}} = -\sum\nolimits_{i=1}^{N} \sum\nolimits_{j=1}^{V} \text{sg}[p_{i,j}^{g}] \log p_{i,j}^{s} \tag{1}$$

where $p_{i,j}^{g}$ and $p_{i,j}^{s}$ respectively represent the probability that the i-th gloss feature $\hat{g}_i$ and sign feature $\hat{s}_i$ are mapped to the j-th virtual token c_j in the codebook, $\text{sg}[\cdot]$ means stop gradient operator. The calculation of $p_{i,j}$ follows the practice of the previous works [19], given a feature $\hat{s}_i$ or $\hat{g}_i$, we apply a linear layer followed by a ReLU and another linear layer to project the feature into logits $l_i \in \mathbb{R}^V$, which denotes the score of each virtual token. During inference, we simply pick the largest index in l_i. During training, we employ Gumbel-Softmax to select discrete tokens in a differentiable manner, the output probability of choosing the j-th token is

$$p_{i,j} = \frac{\exp\left(l_{i,j} + v_j\right)/\tau}{\sum_{k=1}^{V} \exp\left(l_{i,k} + v_k\right)/\tau} \tag{2}$$

where $v = -\log(-\log(u))$ and u are uniform samples from $\mathcal{U}(0,1)$. In the forward pass, we select the token c_k with the highest probability in the codebook to update the original feature $\hat{s}_i$, denoted as $\hat{s}'_i = c_k$, where $k = \text{argmax}_j\, p_{i,j}$. During the backward pass, the gradient is passed to all the Gumbel-Softmax outputs. After that, we can obtain the aligned representations $\hat{\mathbf{S}}' = (\hat{s}'_1, \ldots, \hat{s}'_N) \in \mathbb{R}^{N \times d}$ for $\hat{\mathbf{S}}$. Intuitively, vector quantization can be approximated as mapping sign language representations to gloss representations, thus facilitating to reduce modality gap.

Fusion Module. To enhance the semantic interaction between different modalities, we introduce a fusion module, which consists of a cross-attention layer. Specifically, we

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concatenate the sign representations $\mathbf{S}$ and aligned (or gloss-like) representations $\hat{\mathbf{S}}'$ to obtain the multi-modal features $\mathbf{M}$. Then, $\mathbf{M}$ and $\mathbf{S}$ are fed into the cross-attention layer to output final multi-modal representations $\widetilde{\mathbf{M}}$ which will be input to the text decoder. In calculating the attention, multi-modal features $\mathbf{M}$ are treated as queries, while sign features $\mathbf{S}$ are used as keys and values. The calculation process of the fusion module is as follows:

$$\mathbf{Q} = \mathbf{M} = \left[\mathbf{S}; \hat{\mathbf{S}}'\right] \in \mathbb{R}^{(L_s+N)\times d} \tag{3}$$

$$\mathbf{K} = \mathbf{V} = \mathbf{S} \in \mathbb{R}^{L_s\times d} \tag{4}$$

$$\widetilde{\mathbf{M}} = \mathrm{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \in \mathbb{R}^{(L_s+N)\times d}. \tag{5}$$

Although the gloss is not available during inference, EMF-SLT generates gloss-like features and fuses them with sign features through the multi-modal encoder, thereby facilitating multi-modal semantic interaction in sign language and providing a more informative representation for decoding.

2.2. Multi-task mutual learning

To further bridge the modality gap, we propose a multi-task mutual learning framework that jointly trains the S2T and G2T tasks and regularizes their output distributions to ensure the consistency of decoder outputs across modalities.

Firstly, the multi-modal features $\widetilde{\mathbf{M}}$ and gloss features $\mathbf{G}$ are fed into the text decoder to obtain the output probabilities P_{S2T} and P_{G2T} for the S2T and G2T tasks, respectively. The objective function for both tasks is defined as negative log-likelihood (NLL) loss.

$$\mathcal{L}_{\text{NLL}} = -\sum\nolimits_{i=1}^{T} t_i \ln\left(P^i_{\text{S2T}} \mid t_i\right) - \sum\nolimits_{i=1}^{T} t_i \ln\left(P^i_{\text{G2T}} \mid t_i\right) \tag{6}$$

where t_i and T denote i-th token and the length of the target sentence, respectively.

Then, we use the Kullback–Leibler (KL) divergence [20] to minimize the distance between the output distributions of the two tasks. In this way, the model can dynamically learn and share the knowledge between different modalities, thereby reducing the modality gap. The KL loss is defined as:

$$\mathcal{L}_{\text{KL}} = \mathrm{KL}\left(P_{\text{S2T}} \| P_{\text{G2T}}\right) + \mathrm{KL}\left(P_{\text{G2T}} \| P_{\text{S2T}}\right). \tag{7}$$

In summary, EMF-SLT can be optimized by minimizing the following losses:

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \alpha\mathcal{L}_{\text{KL}} + \beta\mathcal{L}_{\text{Align}}, \tag{8}$$

where α and β are the weights to balance the different losses.

3. EXPERIMENTS

3.1. Experimental Setting

Datasets and Metric. Our model is evaluated on two challenging corpora for SLT, namely, the German sign language dataset (PHOENIX14T) [1] and the Chinese sign language dataset (CSL-Daily) [21]. Following previous studies [4, 21], we use BLEU [22] and ROUGE [23] to measure the translation quality of SLT.

Method	Dev		Test	
	ROUGE	BLEU	ROUGE	BLEU
PHOENIX14T dataset				
SL-Luong [1]	31.8	9.94	31.8	9.58
Joint-SLRT [4]	-	22.38	-	21.32
SignBT [21]	50.29	24.45	49.54	24.32
SignBERT+ [26]	**51.12**	24.95	50.63	25.70
EMF-SLT (Ours)	50.89	**25.4**	**51.72**	**26.01**†
CSL-Daily dataset				
SL-Luong [1]	34.28	7.96	34.54	7.56
Joint-SLRT [4]	37.06	11.88	36.74	11.79
SignBT [21]	49.49	20.80	49.31	21.34
EMF-SLT (Ours)	**52.65**	**22.77**	**51.86**	**22.59**‡

Table 1. Results on PHOENIX14T and CSL-Daily. † and ‡ denote the improvement over our backbone networks is statistically significant ($p < 0.05$).

Implementation Details. We use SentencePiece[2] to learn a joint vocabulary for glosses and texts. The vocabulary sizes of the PHOENIX14T and CSL-Daily datasets are 5k and 4k, respectively. A shared weight matrix is utilized for the input and output word embeddings in the decoder during training. We adopt the Adam optimizer [24] with $\beta_1 = 0.9$ and $\beta_2 = 0.998$, along with the Noam learning rate schedule. Networks are trained using a batch size of 16 and a learning rate of $1e^{-4}$. The dropout rate and label smoothing [25] are set to 0.3 and 0.2, respectively. Setting the temperature τ in Eq.2 to 0.1, and $\alpha = 1$ with $\beta = 0.5$ in Eq.8 for joint training. The codebook size is set to 50.

3.2. Main Results

We evaluate our model against several methods with an identical setup on the PHOENIX14T and CSL-daily datasets, as shown in Table 1. Compared to multi-task learning method Joint-SLRT [4], data augmentation method SignBT [21], and pre-training method SignBERT+ [26], our proposed method achieves better translation quality. In particular, EMF-SLT outperforms these baseline models on PHOENIX14T and CSL-Daily by +0.31 and +1.25 BLEU, respectively. The results demonstrate the effectiveness of our EMF-SLT in improving translation performance.

Our approach manifests a more significant improvement on the CSL-Daily dataset as compared to PHOENIX14T. This differential efficacy is further elucidated upon the computation of the average weightage of shared tokens between gloss and text. Specifically, shared tokens between gloss and text in

[2] https://github.com/google/sentencepiece

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the CSL-Daily test set accounted for 46% of the weight, whereas the corresponding weight in the PHOENIX14T dataset amounted to a meager 12%. The superior performance on the CSL-Daily dataset supports our initial intuition that bridging the modality gap via alignment and fusion of gloss and sign representations tends to yield more substantial improvements when the gloss and text exhibit higher linguistic consistency.

3.3. Ablation Study

To further comprehend the contributions of our proposed multi-modal encoder and multi-task mutual learning method, we conduct ablation studies on the PHOENIX14T and CSL-Daily test set. As shown in Table 2, each component of the EMF-SLT is indispensable, contributing positively to the overall performance. For PHOENIX14T and CSL-Daily, the multi-modal encoder brings 0.71 BLEU and 2.53 BLEU improvements, while the multi-task mutual learning achieves gains of 1.38 BLEU and 0.52 BLEU, respectively.

To test whether our approach mitigates the modality gap, we first recalculate the average sentence-level L2 distance measurements between sign videos and texts on the PHOENIX14T test set. Our method achieves a significant reduction in the L2 distance with a score of 6.89, compared to a score of 15.26 for the baseline[4].

Furthermore, we also analyze token-level representation discrepancy. To be concrete, we employ the CTC classifier of pre-trained SMKD [17] as a sign-gloss aligner to obtain approximate token-level alignment. Given the gloss token representation g_i and its corresponding sign segment representation $[m_i, \ldots, m_{i+l}]$, we use a mean-pooling operation to extract the token-level sign representation a_i, where l denotes the length of a segment. We then calculate the average cosine similarity across all respective a_i and g_i within the test set. As illustrated in Table 3, EMF-SLT significantly enhances the similarity between gloss and sign representations. This can be attributed to the alignment and fusion performed by our proposed multi-modal encoder. Notably, when all components of the EMF-SLT are removed, the cross-modal similarity in the CSL-Daily dataset decreases significantly compared to the PHOENIX14T dataset, along with a significant decline in the BLEU score. This demonstrates that EMF-SLT can effectively bridge the representation gap across modalities, thereby improving the performance of SLT.

Model	PHOENIX14T	CSL-Daily
EMF-SLT	**26.01**	**22.59**
w/o KL loss	24.63 (-1.38)	22.07 (-0.52)
w/o Fusion Module	24.27 (-1.74)	20.98 (-1.61)
w/o Align loss	23.92 (-2.09)	19.54 (-3.05)

Table 2. Ablation study of contribution for each component in PHOENIX14T and CSL-Daily test set.

Model	PHOENIX14T	CSL-Daily
EMF-SLT	**40.31**	**62.33**
w/o all components	29.15	33.86

Table 3. Comparison of token-level representation similarity across modalities.

4. RELATED WORKS

Camgoz et al [1] first introduced an end-to-end neural S2T model that integrates Convolutional Neural Networks and the attention-based sequence-to-sequence model. However, the performance of the end-to-end method is heavily dependent on large-scale and high-quality parallel data. Many techniques, such as multi-task learning [4, 6, 8], data augmentation [21], and pretraining [5, 7, 26] are applied to utilize the data from related tasks to improve the sample efficiency and reduce the modality gap in low-resource scenarios.

A typical article using multi-task learning like [4] introduced a novel framework that employs a multi-task supervision mechanism with gloss to remediate the modality gap, while simultaneously learning S2G and S2T jointly. Zhou et al [21] proposed a two-stage sign back-translation methodology, consisting of text-to-gloss and gloss-to-sign stages, for producing synthetic data to augment S2T training. Hu et al. [26] proposed a self-supervised pre-trainable model with hand gestures, utilizing a model-aware hand prior for understanding sign language and mitigating over-fitting by using multiple levels of masked modeling strategies.

Contrastingly, our methodology explicitly fuses gloss and sign modalities representations, an approach tailored towards mitigating the modality gap. This pivotal aspect was not given due consideration in previous studies, highlighting the innovative nature of our approach.

5. CONCLUSTION

In this paper, we present EMF-SLT, a novel end-to-end sign language translation model that explicitly fuses sign features and gloss-like features to bridge the gap between modalities. Furthermore, EMF-SLT employs a mutual learning training paradigm designed to enhance performance through the knowledge interaction of S2T and G2T components. Experimental evaluations on the challenging PHOENIX14T and CSL-Daily benchmark datasets demonstrate the effectiveness of our approach for sign language translation.

6. ACKNOWLEDGEMENTS

We would like to thank all the anonymous reviewers for their insightful comments. This work was supported by National Natural Science Foundation of China (No. 62076211).

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